

1. $A = x^2 + 3x + 5$, $B = x^2 - 3x - 3$, $C = x^2 - 2x + 4$ であるとき、次の計算をせよ。

(1) $-B - C$

答. _____

(2) $A - 3B$

答. _____

(3) $A + B + C$

答. _____

(4) $-A + B - C$

答. _____

(5) $3A - 2B - 3C$

答. _____

(6) $A + 3B - C$

答. _____

(7) $A - 3B + C$

答. _____

(8) $3A + 3B - C$

答. _____

2. 次の1次方程式を解け。

(1) $-x - 6 = 3x + 6$

答. _____

(2) $7x - 7 = 9x + 4$

答. _____

(3) $-5x - 1 + 2(-9x - 3) = 0$

答. _____

(4) $8x - 3(6x - 4) = -2$

答. _____

(5) $7 - 3(-2x - 2) = -5x$

答. _____

(6) $-4x - \frac{1}{3} = \frac{1}{3}x + 2$

答. _____

(7) $\frac{9}{4}x - 5 = \frac{4}{3}x + \frac{8}{3}$

答. _____

(8) $-\frac{9}{2}x - \frac{9}{2} = 2x + \frac{3}{2}$

答. _____

1. $A = x^2 + 3x + 5, B = x^2 - 3x - 3, C = x^2 - 2x + 4$ であるとき、次の計算をせよ。

(1) $-B - C$

答. $-2x^2 + 5x - 1$

(2) $A - 3B$

答. $-2x^2 + 12x + 14$

(3) $A + B + C$

答. $3x^2 - 2x + 6$

(4) $-A + B - C$

答. $-x^2 - 4x - 12$

(5) $3A - 2B - 3C$

答. $-2x^2 + 21x + 9$

(6) $A + 3B - C$

答. $3x^2 - 4x - 8$

(7) $A - 3B + C$

答. $-x^2 + 10x + 18$

(8) $3A + 3B - C$

答. $5x^2 + 2x + 2$

2. 次の 1 次方程式を解け。

(1) $-x - 6 = 3x + 6$

$-4x = 12$

答. $x = -3$

(2) $7x - 7 = 9x + 4$

$-2x = 11$

答. $x = -\frac{11}{2}$

(3) $-5x - 1 + 2(-9x - 3) = 0$

$-5x - 1 - 18x - 6 = 0$

$-23x = 7$

答. $x = -\frac{7}{23}$

(4) $8x - 3(6x - 4) = -2$

$8x - 18x + 12 = -2$

$-10x = -14$

答. $x = \frac{7}{5}$

(5) $7 - 3(-2x - 2) = -5x$

$7 + 6x + 6 = -5x$

$11x = -13$

答. $x = -\frac{13}{11}$

(6) $-4x - \frac{1}{3} = \frac{1}{3}x + 2$

両辺に 3 をかけて、 $-12x - 1 = x + 6$

答. $x = -\frac{7}{13}$

(7) $\frac{9}{4}x - 5 = \frac{4}{3}x + \frac{8}{3}$

両辺に 12 をかけて、 $27x - 60 = 16x + 32$

答. $x = \frac{92}{11}$

(8) $-\frac{9}{2}x - \frac{9}{2} = 2x + \frac{3}{2}$

両辺に 2 をかけて、 $-9x - 9 = 4x + 3$

答. $x = -\frac{12}{13}$